ResQ-MARL: URBAN SEARCH & RESCUE DIGITAL TWIN

An Autonomous Multi-Agent RL Simulation for Post-Earthquake Survivor Extraction

Target Domain: Disaster Robotics & USAR | **Framework:** Unity ML-Agents & OpenXR | **Status:** Final Review Proposal

Executive Summary & Academic Value:
*Post-earthquake search and rescue in structurally collapsed buildings poses extreme risks to human responders. This project pivots our Multi-Agent Reinforcement Learning engine into **ResQ-MARL**: an interactive simulation and digital twin testbed that trains autonomous robot swarms (UAVs/UGVs) to navigate rubble, detect trapped victims using thermal sensors, clear passage corridors, and guide survivors back to a designated Safe Triage Zone. It incorporates a **Virtual Incident Command Station** with hand-tracking/VR to inspect 3D damage and map real-world facilities.*

1. Direct System & Codebase Mapping (Game to Disaster Testbed)

All existing subsystems map seamlessly to real-world disaster robotics functions without starting from scratch:

Legacy Game Element	ResQ-MARL Disaster Function	Technical & Real-World Justification
Seekers (AI Agents)	**Autonomous Rescue UAV/UGV Swarm**	Autonomous drones & robotic crawlers trained via PPO to cooperatively sweep occluded ruins.
Hiders (AI Agents)	**Trapped Earthquake Victims**	Survivors sheltering under structural pockets awaiting location and extraction.
Movable Crates / Blocks	**Structural Debris & Fallen Rubble**	Dynamic obstacles cleared using robotic manipulators (BoxHolding.cs) to open passage corridors.
Laser Scanners	**LiDAR & Thermal Life-Sign Sensors**	Infrared line-of-sight sensors detecting survivor thermal signatures through rubble crevices.
Architect Mode & Layouts	**Incident Command & Digital Twin Mapper**	Coordinators map real-world building blueprints (PlayerLayout.cs) to run pre-mission sweep simulations.

2. Virtual Incident Command Station (VR & Hand Tracking)

Directly addressing the review committee's feedback regarding hands-on interaction and VR immersion:

- **Natural Gesture Debris Manipulation:** Using a standard laptop webcam with MediaPipe (or VR hand controllers), the human disaster coordinator uses pinch and grasp gestures to pick up, clear, and stabilize virtual debris in real time.
- **Hazard Tagging & Waypointing:** The operator points to mark structural collapse zones or assign tactical patrol waypoints for the autonomous swarm to prioritize.
- **Immersive 1:1 Scale Damage Survey:** Responders can enter a first-person VR perspective to inspect structural damage safely before human teams enter the physical site.

3. Safe Triage Zone & Real-World Digital Twin Workflow

The platform introduces two vital end-to-end operational capabilities:

- **Victim Extraction to Safe Triage Base:** Upon locating a victim, the rescue robot establishes an evacuation corridor and guides the survivor back to a designated green Triage Base, shifting the mission objective from simple detection to complete survival extraction.

- **Real-Life Facility Import (Sim-to-Real):** By importing floor plans via JSON (already implemented in PlayerLayout.cs), rescue teams can digitize real earthquake-damaged facilities (schools, hospitals, offices) and evaluate multiple swarm search policies prior to deployment.

4. Quantitative Mission & Safety Evaluation Metrics

The system replaces gaming scores with objective disaster response metrics exported to CSV:

Metric	Definition	Operational Impact
Mean Time to Rescue (MTTR)	Average elapsed seconds to detect and extract survivors.	Critical for maximizing survival within the 72-hour Golden Window.
Swarm Coverage Rate (%)	Percentage of the building surveyed by thermal sensors.	Ensures zero uninspected rooms or dead zones remain.
Blind-Spot Void Density	Unexplored rubble pockets displayed on the floor heatmap.	Directs human secondary search teams to high-risk shadowed voids.
Debris Obstruction Factor	Delay inflicted by collapsed beams and rubble.	Evaluates whether robotic clearing or alternative routing is optimal.

5. The 5-Minute Professor Review Script

A structured oral and visual walkthrough engineered for maximum marks during the final defense:

Minute 1: The Problem & Humanitarian Need: Frame post-earthquake structural hazards and introduce ResQ-MARL as an autonomous multi-agent simulation testbed and digital twin.

Minute 2: Incident Command & Debris Setup: Load the 'Collapsed Building' scenario. Demonstrate real-time debris manipulation and hazard zone marking.

Minute 3: Thermal / LiDAR Swarm Sweep: Launch simulation. Press 'H' to reveal the live floor heatmap demonstrating dynamic thermal coverage and dead-zone identification.

Minute 4: Multi-Agent Policy Comparison: Compare naive baseline search against our 200M self-play PPO policy, showing emergent sector division and cooperative exploration.

Minute 5: Extraction to Safe Base & Telemetry Export: Guide victim into the Safe Triage Base. Click 'Export Audit Report' to display the generated CSV dossier containing MTTR and coverage statistics.